

WHITEBRIDGE HIGH SCHOOL



HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION

2005
1999

MATHEMATICS

4 UNIT (ADDITIONAL)

Time Allowed: Three hours
(Plus 5 minutes reading time)

Directions to Candidates

- Attempt all questions
- ALL questions are of equal value
- All necessary working should be shown. Marks may be deducted for careless or badly arranged work.
- Standard integrals are provided
- Board-approved calculators may be used
- Each question is to be returned on a separate sheet of paper clearly labelled, showing your Name and Student Number.

Question 1

(mark)

2 a) Find $\int \frac{e^x dx}{e^{2x} + 1}$

2 b) Find $\int x^2 \cos x \, dx$

3 c) Evaluate $\int_0^{\pi/2} \frac{dx}{2 + \cos x}$

4 d) Evaluate $\int_0^1 \frac{3x^2 - 2x + 1}{(x^2 + 1)(x^2 + 2)} dx$

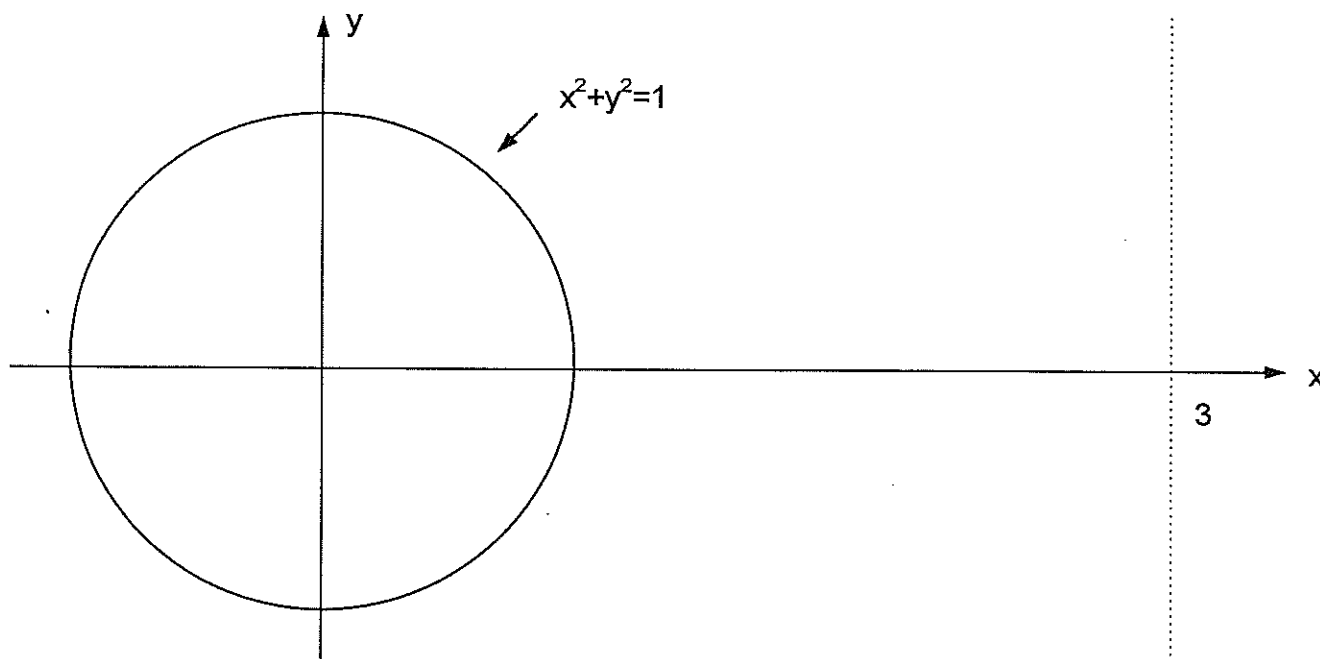
4 e) Find $\int \frac{x+3}{\sqrt{x^2 - 2x + 5}} dx$

15

Question 2

a) 2 (i) Express $-1 + \sqrt{3}i$ in Modulus-Argument form.2 (ii) Hence evaluate $(-1 + \sqrt{3}i)^6$ 4 b) Find the complex 5th roots of -1 .3 c) Sketch the regions where the inequalities $|z - 3 + i| \leq 5$ and $|z + 1| \leq |z - 1|$ both hold.4 d) If $\text{Arg}(z - 2) = \text{Arg}(z + 2) + \frac{\pi}{3}$, show that the locus of the point P representing z on an Argand diagram is an arc of a circle and find the centre and radius of this circle.

Question 3



The circle $x^2 + y^2 = 1$ is rotated about the line $x = 3$ to form a ring (Torus).

Find the volume of the ring by

- 8 a) the slice technique,
7 b) the method of cylindrical shells.

Question 4

- 3 a) The polynomial $z^3 - 7z^2 + 25z - 39$ has one zero equal to $2 + 3i$. Write down its three linear factors.
- 4 b) If $P(x) = x^4 - 3x^3 - 6x^2 + 28x - 24$ has a triple zero, find all the zeros and factor $P(x)$ over the real numbers.
- 4 c) The equation $x^4 - px^3 + qx^2 - pqx + 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Show that $(\alpha + \beta + \gamma)(\alpha + \beta + \delta)(\alpha + \gamma + \delta)(\beta + \gamma + \delta) = 1$
- 4 d) α, β, γ are the roots of $x^3 + qx + r = 0$. Find in terms of q, r the equation with roots $\frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2}$

Question 5

- a) Let $f(x) = -x^2 + 5x - 4$

On separate diagrams and without using calculus, sketch the following graphs. Indicate clearly any asymptotes and intercepts with the axes.

1 (i) $y = f(x)$

2 (ii) $y = |f(x)|$

2 (iii) $y^2 = f(x)$

2 (iv) $y = \frac{1}{f(x)}$

2 (v) $y = e^{f(x)}$

- b) Solve the following inequality with the aid of an appropriate sketch.

3 $|x - 1| + |x + 1| > 3$

- c) Sketch $y = x \cos x$ where $-2\pi \leq x \leq 2\pi$

3
15

Question 6

- a) For the hyperbola $\frac{x^2}{4} - \frac{y^2}{12} = 1$

Find (i) the coordinates of the foci

(ii) the equations of the directrices

(iii) the equations of the asymptotes

3 (iv) the equation of the tangent at $P(2 \sec \theta, 2\sqrt{3} \tan \theta)$

3 (v) given that $0 < \theta < \pi/2$ show that the point of intersection of the above tangent and the nearer directrix has y -coordinate $\frac{6 - 12 \cos \theta}{2\sqrt{3} \sin \theta}$

Question 6 continued

- b) $P(ct, \frac{c}{t})$ lies on the rectangular hyperbola $xy = c^2$. The normal at P meets the rectangular hyperbola $x^2 - y^2 = a^2$ at Q and R.

3 (i) Find the equation of the normal at P

3 (ii) Show that P is the midpoint of QR

3
15

Question 7

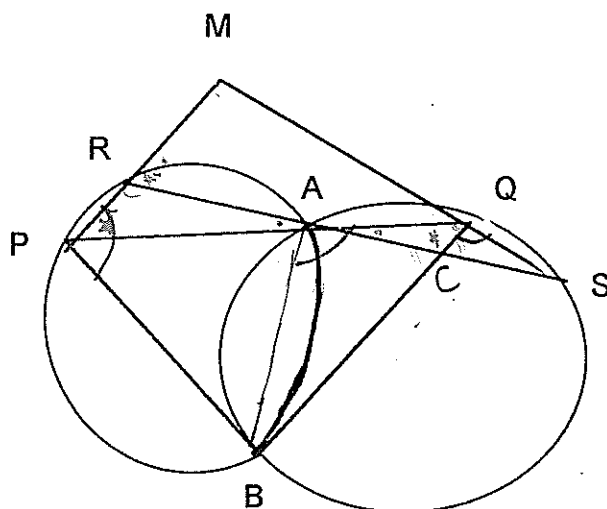
- a) If x, y, z are positive real numbers, prove the following:

1 (i) $x^2 + y^2 \geq 2xy$

2 (ii) $\frac{x}{y} + \frac{y}{x} \geq 2$

4 (iii) $x^3 + y^3 + z^3 \geq 3xyz$

- b)



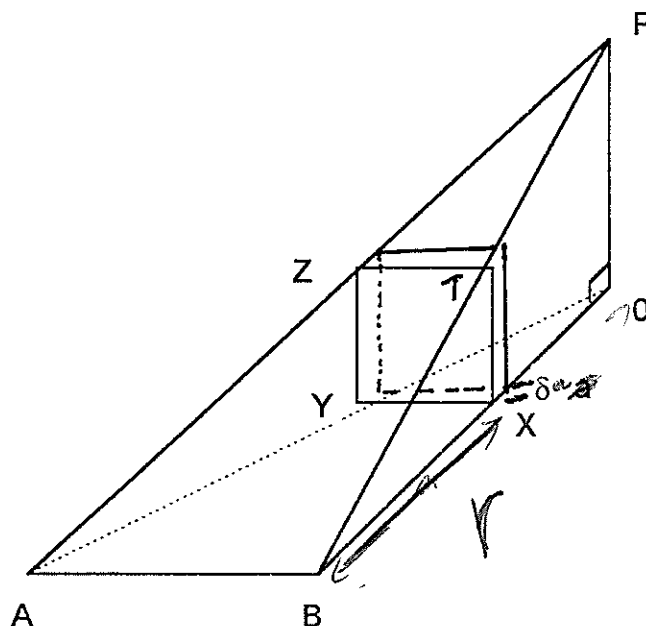
Two circles ARPB, AQS B intersect at A & B. PAQ and RAS are straight lines. PR and SQ are produced to meet at M.

Prove that MPBQ is a cyclic quadrilateral.

4 c) Prove that $\cos^6 \theta - \sin^6 \theta = \cos 2\theta \left(1 - \frac{1}{4} \sin^2 2\theta \right)$.

Question 8

a)



Let ABO be an isosceles triangle. $AO = BO = r$, $AB = b$

Let PABO be a triangular pyramid with height $OP = h$ and OP perpendicular to the plane of ABO as in the diagram.

Consider a slice S of the pyramid of width δa as in the diagram.

The slice S is perpendicular to the plane of ABO at XY with $XY \parallel AB$ and $XB = a$.

Note $XT \parallel OP$.

(i) Show that the volume of S is $(\frac{r-a}{r})b(\frac{ah}{r})\delta a$. When δa is small.

(You may assume that the slice is approximately a rectangular prism of base XYZT and height δa)

(ii) Hence show that the pyramid PABO has volume $\frac{1}{6}hbr$.

b) For $n = 1, 2, 3, \dots$, let $S_n = 1 + \sum_{r=1}^n \frac{1}{r!}$

(i) Prove by mathematical induction that $e - S_n = e \int_0^1 \frac{x^n}{n!} e^{-x} dx$

Question 8 continued

(ii) From (i) deduce that $0 < e - S_n < \frac{3}{(n+1)!}$ for $n = 1, 2, 3, \dots$

2

(N.B. $e < 3$ and $e^{-x} \leq 1$ for $x \geq 0$)

15

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

1999 4-UNIT TRIAL SOLUTIONS

WHITEBRIDGE HIGH

QUESTION 1:

$$(a) \int \frac{e^x dx}{e^{2x} + 1}$$

$$\text{Let } u = e^x \\ \therefore \frac{du}{dx} = e^x \\ du = e^x dx$$

$$= \int \frac{du}{u^2 + 1}$$

$$= \tan^{-1} u$$

$$= \tan^{-1} e^x + C$$

$$(d) \int_0^1 \frac{3x^2 - 2x + 1}{(x^2 + 1)(x^2 + 2)} dx$$

$$\frac{3x^2 - 2x + 1}{(x^2 + 1)(x^2 + 2)} = \frac{ax + b}{x^2 + 1} + \frac{cx + d}{x^2 + 2}$$

$$= \frac{ax^3 + bx^2 + 2ax + 2b + cx^3 + dx^2 + cx + d}{(x^2 + 1)(x^2 + 2)}$$

$$= \frac{(a+c)x^3 + (b+d)x^2 + (2a+c)x + 2b+d}{(x^2 + 1)(x^2 + 2)}$$

$$\begin{cases} a + c = 0 \\ b + d = 3 \\ 2a + c = -2 \\ 2b + d = 1 \end{cases} \Rightarrow \begin{cases} a = -2 \\ c = 2 \\ b = -2 \\ d = 5 \end{cases}$$

$$\therefore \int_0^1 \frac{-2x - 2}{x^2 + 1} + \frac{2x + 5}{x^2 + 2} dx$$

$$= -\int_0^1 \frac{2x}{x^2 + 1} - \int_0^1 \frac{2}{1 + x^2} + \int_0^1 \frac{2x}{x^2 + 2} + \int_0^1 \frac{5}{x^2 + 2} dx$$

$$= \left[-\ln(x^2 + 1) - 2 \tan^{-1} x + \ln(x^2 + 2) + \frac{5}{\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} \right]_0^1$$

$$= \left(-\ln 2 - 2 \tan^{-1} 1 + \ln 3 + \frac{5}{\sqrt{2}} \tan^{-1} \frac{1}{\sqrt{2}} \right) - \left(-\ln 1 - 2 \tan^{-1} 0 + \ln 2 + \frac{5}{\sqrt{2}} \tan^{-1} 0 \right)$$

$$= -\ln 2 - \pi/2 + \ln 3 + \frac{5}{\sqrt{2}} \tan^{-1} \frac{1}{\sqrt{2}} - \ln 2$$

$$= \ln 3 - 2 \ln 2 - \pi/2 + \frac{5}{\sqrt{2}} \tan^{-1} \frac{1}{\sqrt{2}}$$

$$(b) \int x^2 \cos x = x^2 \sin x - \int 2x \sin x \\ = x^2 \sin x - 2 \left\{ -x \cos x - \int -\cos x \times 1 \right\} \\ = x^2 \sin x + 2x \cos x - 2 \sin x + C$$

$$(c) \int_0^{\pi/2} \frac{dx}{2 + \cos x}$$

$$\text{LET } t = \tan \frac{x}{2}$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$= \frac{1}{2} (1 + t^2)$$

$$\therefore dx = \frac{2 dt}{1 + t^2}$$

$$= \int_0^1 \frac{2 dt}{1 + t^2} \div 2 + \frac{1 - t^2}{1 + t^2}$$

$$= \int_0^1 \frac{2}{1 + t^2} \times \frac{1 + t^2}{3 + t^2} dt$$

$$= \int_0^1 \frac{2}{3 + t^2} dt$$

$$= \frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1 = \frac{2}{\sqrt{3}} \left\{ \tan^{-1} \frac{1}{\sqrt{3}} - \tan^{-1} 0 \right\}$$

$$= \frac{2}{\sqrt{3}} \times \frac{\pi}{6}$$

$$= \frac{\pi}{3\sqrt{3}}$$

$$(e) \int \frac{x+3}{\sqrt{x^2-2x+5}} dx = \int \frac{x-1+4}{\sqrt{x^2-2x+5}} dx$$

$$= \int \frac{x-1}{\sqrt{x^2-2x+5}} + \int \frac{4}{\sqrt{x^2-2x+5}} dx$$

$$\int \frac{x-1}{\sqrt{x^2-2x+5}} dx \quad \text{Let } u = x^2-2x$$

$$\frac{du}{dx} = 2x-2$$

$$= 2(x-1)$$

$$\therefore \frac{du}{2} = (x-1) dx$$

$$= \frac{1}{2} \int \frac{du}{\sqrt{u+5}}$$

$$= \frac{1}{2} \int (u+5)^{-1/2} du$$

$$= \frac{1}{2} \times 2 (u+5)^{1/2}$$

$$= \sqrt{u+5}$$

$$= \sqrt{x^2-2x+5}$$

$$\int \frac{4}{\sqrt{x^2-2x+5}} = \int \frac{4}{\sqrt{x^2-2x+1+4}}$$

$$= \int \frac{4}{\sqrt{(x-1)^2+4}} dx$$

$$= 4 \ln \left[(x-1) + \sqrt{(x-1)^2+4} \right]$$

$$\therefore \int \frac{x+3}{\sqrt{x^2-2x+5}} = \sqrt{x^2-2x+5} + 4 \ln \left[(x-1) + \sqrt{x^2-2x+5} \right] + C$$

Question 2: 3

$$(a) \quad \perp \quad |-1 + \sqrt{3}i| = \sqrt{1^2 + (\sqrt{3})^2} = 2$$

$$\text{ARG}(-1 + \sqrt{3}i) = \theta \quad \text{WHERE} \quad \tan \theta = -\sqrt{3}$$

$$\therefore \theta = \frac{2\pi}{3}$$

$$\therefore -1 + \sqrt{3}i = 2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$$

$$(ii) \quad (-1 + \sqrt{3}i)^6 = \left[2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \right]^6$$

$$= 2^6 \left[\cos 4\pi + i \sin 4\pi \right]$$

$$= 2^6 [1 + 0]$$

$$= 64$$

$$(b) \quad \text{LET } z^5 = -1$$

$$\text{Let } z = r(\cos \theta + i \sin \theta)$$

$$\therefore r^5 (\cos 5\theta + i \sin 5\theta) = 1(\cos \pi + i \sin \pi)$$

$$\therefore r^5 = 1$$

$$\therefore r = 1$$

$$5\theta = \pi + 2k\pi$$

$$\therefore \theta = \frac{\pi}{5} + \frac{2k\pi}{5}$$

$$\theta_1 = \frac{\pi}{5} + \frac{2\pi}{5} = \frac{3\pi}{5}$$

$$\theta_2 = \frac{\pi}{5} - \frac{2\pi}{5} = -\frac{\pi}{5}$$

$$\theta_3 = \frac{\pi}{5}$$

$$\theta_4 = \frac{\pi}{5} + \frac{4\pi}{5} = \pi$$

$$\theta_5 = \frac{\pi}{5} - \frac{4\pi}{5} = -\frac{3\pi}{5}$$

$$\therefore z_1 = \left(\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5} \right)$$

$$\therefore z_2 = \left(\cos \frac{-\pi}{5} + i \sin \frac{-\pi}{5} \right)$$

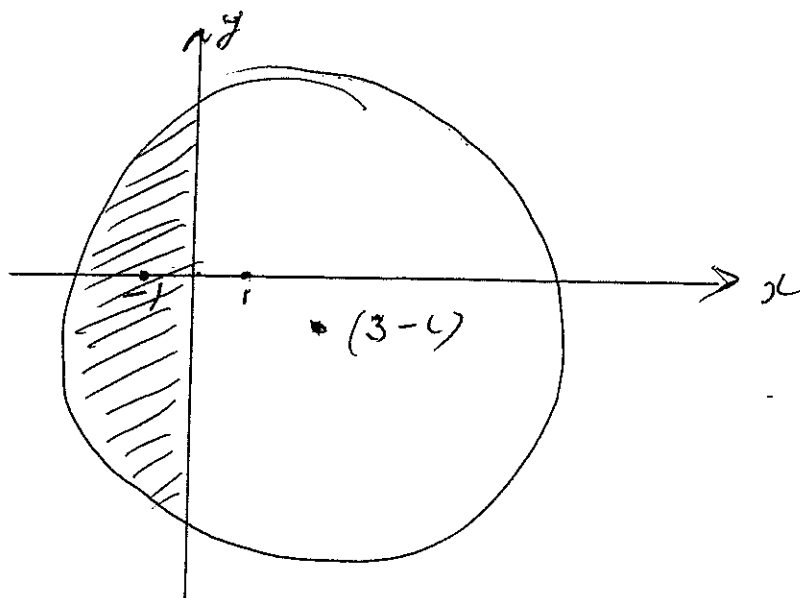
$$\therefore z_3 = \left(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5} \right)$$

$$\therefore z_4 = \left(\cos \pi + i \sin \pi \right)$$

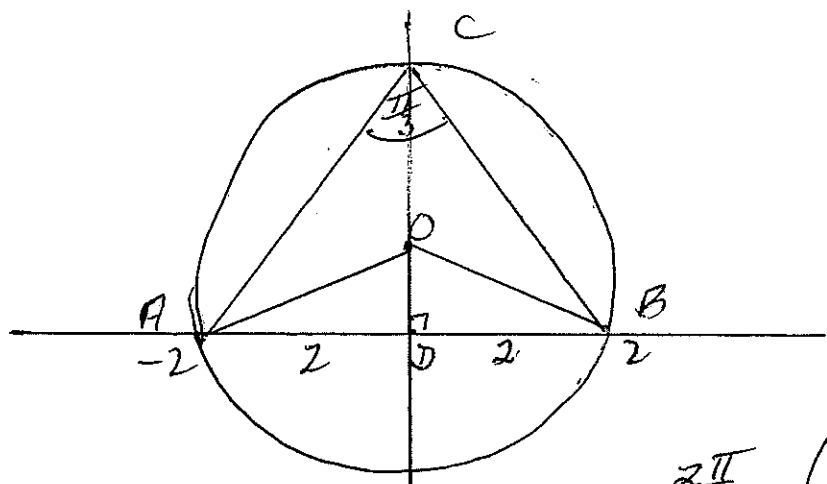
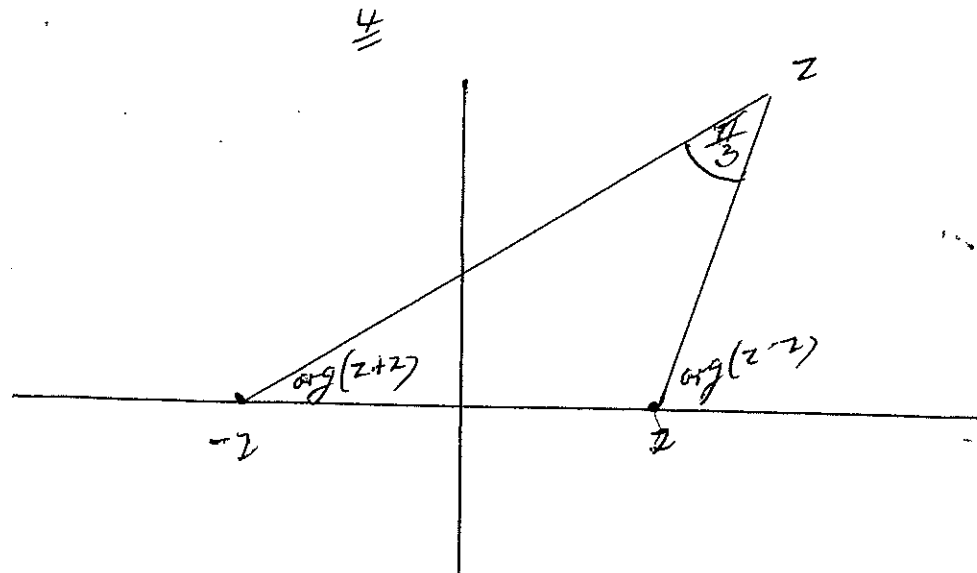
$$\therefore z_5 = \left(\cos \frac{-3\pi}{5} + i \sin \frac{-3\pi}{5} \right)$$

- if $k=1$:
- if $k=-1$:
- if $k=0$:
- if $k=2$:
- if $k=-2$:

(c)



(d)



ANGLE AT CENTRE = $\frac{2\pi}{3}$

$$\begin{aligned} \therefore \angle AOB &= \frac{2\pi}{3} \\ \therefore \tan \frac{\pi}{3} &= \frac{OD}{2} \\ \therefore OD &= \frac{2}{\sqrt{3}} \end{aligned}$$

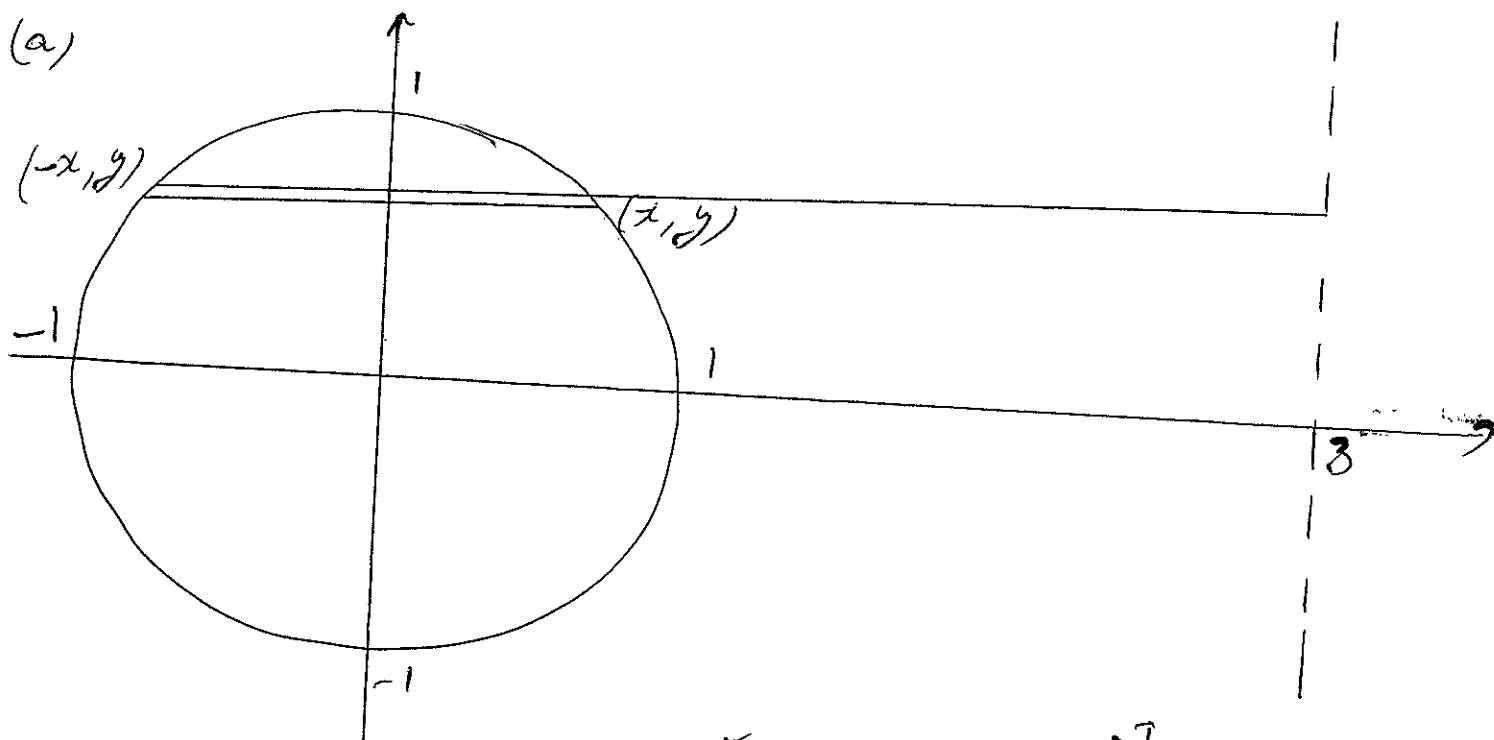
$\therefore$ CENTRE $(0, \frac{2}{\sqrt{3}})$

$$\begin{aligned} \text{RADIUS} &= \sqrt{(2-0)^2 + (0-\frac{2}{\sqrt{3}})^2} \\ &= \sqrt{4 + \frac{4}{3}} \\ &= \sqrt{\frac{16}{3}} = \frac{4}{\sqrt{3}} \end{aligned}$$

(AB = CHORD SUBTENDING A CONSTANT ANGLE $\frac{\pi}{3}$ i.e. ON CIRCUMFERENCE OF CIRCLE)

~~EQUATION~~

QUESTION 3: 5



VOLUME
AREA OF SLICE = $\left[\pi (3+x)^2 - \pi (3-x)^2 \right] x \delta y$

$$\therefore V = \pi \int_{-1}^1 9 + x^2 + 6x - 9 - x^2 + 6x$$

$$= \pi \int_{-1}^1 12x \, dy$$

$$= 12\pi \int_{-1}^1 \sqrt{1-y^2} \, dy$$

Let $y = \sin \theta$

$$\frac{dy}{d\theta} = \cos \theta$$

$$dy = \cos \theta \, d\theta$$

$$= 12\pi \int_{-\pi/2}^{\pi/2} \sqrt{1-\sin^2 \theta} \cos \theta \, d\theta$$

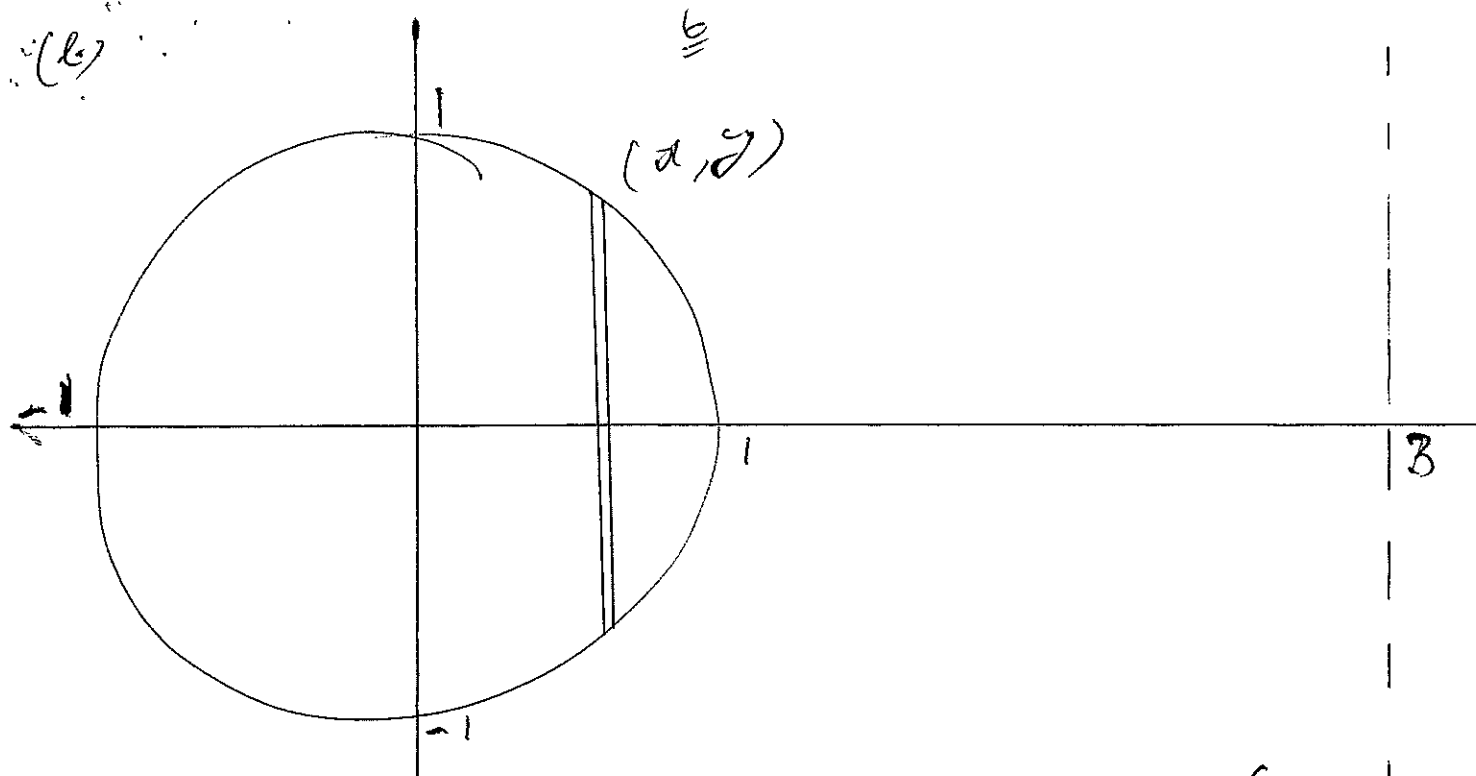
$$= 12\pi \int_{-\pi/2}^{\pi/2} \cos^2 \theta \, d\theta$$

$$= 12\pi \int_{-\pi/2}^{\pi/2} \frac{1}{2} (1 + \cos 2\theta) \, d\theta$$

$$= 6\pi \left[\theta + \frac{\sin 2\theta}{2} \right]_{-\pi/2}^{\pi/2}$$

$$= 6\pi \left[\left(\frac{\pi}{2} + \frac{\sin \pi}{2} \right) - \left(-\frac{\pi}{2} + \frac{\sin(-\pi)}{2} \right) \right]$$

$$= 6\pi \times \pi = 6\pi^2 \text{ units}^3$$



VOLUME
~~AREA~~ OF SLICE = $2\pi(3-x) \times 2y \, dx$

$$V = \int_{-1}^1 2\pi(3-x) \times 2y \, dx$$

$$= 4\pi \int_{-1}^1 (3-x) \sqrt{1-x^2} \, dx$$

$$= 12\pi \int_{-1}^1 \sqrt{1-x^2} \, dx - 4\pi \int_{-1}^1 x \sqrt{1-x^2} \, dx$$

$$= 6\pi^2 \quad \text{--- Let } u = 1-x^2$$

(FROM PART (a))

$$\frac{du}{dx} = -2x$$

$$\therefore -\frac{du}{2} = x \, dx$$

$$\therefore -4\pi \int_{-1}^1 x \sqrt{1-x^2} \, dx = +\frac{4\pi}{2} \int_0^0 \sqrt{u} \, du$$

= No solution

$$\therefore V = 6\pi^2 \text{ units}^3$$

QUESTION 4:

(a) If $2+3i$ is a root then $2-3i$ is also a root as polynomial has real coefficients.

$\therefore$ Roots are $2+3i$, $2-3i$ & α

$$\text{Sum of roots} = 2+3i + 2-3i + \alpha = 7$$

$$\therefore \alpha = 3$$

$$\therefore \text{Factors are } [x-(2+3i)][x-(2-3i)][x-3]$$
$$= (x-2-3i)(x-2+3i)(x-3)$$

(b) If $P(x) = x^4 - 3x^3 - 6x^2 + 28x - 24$ has a triple root

$$P'(x) = 4x^3 - 9x^2 - 12x + 28$$

has a double root

$$P''(x) = 12x^2 - 18x - 12$$

has a single root

$$\therefore \begin{aligned} 12x^2 - 18x - 12 &= 0 \\ 2x^2 - 3x - 2 &= 0 \end{aligned}$$

$$(2x+1)(x-2) = 0$$
$$x = 2, -\frac{1}{2}$$

$P'(2) = 0 \therefore x=2$ is the triple root

$\therefore$ Roots are $2, 2, 2, \alpha$

$$\text{Sum of roots} = 6 + \alpha = 3$$
$$\alpha = -3$$

$\therefore$ Zeros are $2, 2, 2, -3$

$$\text{Factors are } (x-2)^3(x+3)$$

$$(4) (d) \quad x^3 + 9x + 5 = 0$$

$$\alpha + \beta + \gamma = 0$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = 9$$

$$\alpha\beta\gamma = -5$$

$$\begin{aligned} \text{Sum of roots} &= \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} \\ &= \frac{\beta^2\gamma^2 + \alpha^2\gamma^2 + \alpha^2\beta^2}{\alpha^2\beta^2\gamma^2} \\ &= \frac{(\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)}{(\alpha\beta\gamma)^2} \end{aligned}$$

$$= \frac{9 + 2 \times 0}{(-5)^2}$$

$$= \frac{9}{25}$$

Double

$$= \frac{\frac{1}{\alpha^2\beta^2} + \frac{1}{\alpha^2\gamma^2} + \frac{1}{\beta^2\gamma^2}}{\gamma^2 + \alpha^2 + \beta^2}$$

$$= \frac{(\alpha + \beta + \gamma)^2 - 2(\alpha\gamma + \beta\alpha + \beta\gamma)}{(\alpha\beta\gamma)^2}$$

$$= \frac{0 - 2 \times 9}{25}$$

$$= -\frac{18}{25}$$

For the

$$= \frac{1}{\alpha^2\beta^2\gamma^2}$$

$$= \frac{1}{(-5)^2}$$

$$= \frac{1}{25}$$

$$\therefore x^3 - \frac{9}{25}x^2 - \frac{18}{25}x - \frac{1}{25} = 0$$

$$25x^3 - 9x^2 - 18x - 1 = 0$$

$$(c) \cdot x^4 - px^3 + qx^2 - pqx + 1 = 0 \quad //$$

$$\alpha + \beta + \gamma + \delta = p$$

$$\therefore \alpha + \beta + \gamma = p - \delta$$

$$\begin{aligned} \therefore (\alpha + \beta + \gamma)(\alpha + \beta + \delta)(\alpha + \gamma + \delta)(\beta + \gamma + \delta) &= (p - \delta)(p - \delta)(p - \delta)(p - \delta) \\ &= (p^2 - p(\delta + \delta) + \delta\delta)(p^2 - p(\delta + \delta) + \delta\delta) \\ &= p^4 - p^3(\delta + \delta) + p^2\delta\delta - p^3(\alpha + \beta) + p^2(\delta + \delta)(\alpha + \beta) - p(\alpha\delta\delta + \delta\delta\beta) \\ &\quad + p^2\alpha\beta - p(\alpha\beta\delta + \alpha\beta\delta) + 2\beta\delta\delta \\ &= p^4 - p^3(\alpha + \beta + \delta + \delta) + p^2(\delta\delta + 2\delta + \beta\delta + 2\delta + \beta\delta + \alpha\beta) \\ &\quad - p(\alpha\delta\delta + \delta\delta\beta + 2\beta\delta + 2\beta\delta) + 2\beta\delta\delta \\ &= p^4 - p^3 \times p + p^2 \times 2 - p \times pq + 1 \\ &= p^4 - p^4 + p^2q - p^2q + 1 \\ &= 1 \\ &= R.H.S. \end{aligned}$$

(d) Let $x = \alpha, \beta, \gamma$ ARE ROOTS OF $x^3 + qx + r = 0$
 Let $y = \frac{1}{x^2}$ TO REPRESENT ROOTS $\frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2}$
 $\therefore x^2 = \frac{1}{y}$
 $x = \frac{1}{\sqrt{y}}$

$$\therefore \left(\frac{1}{\sqrt{y}}\right)^3 + q\left(\frac{1}{\sqrt{y}}\right) + r = 0$$

$$\frac{1}{y\sqrt{y}} + \frac{q}{\sqrt{y}} + r = 0$$

$$\therefore \left(\frac{1}{y\sqrt{y}} + \frac{q}{\sqrt{y}}\right)^2 = -r$$

$$\therefore \frac{1}{y^3} + \frac{q}{y} + \frac{2q}{y^2} = r^2$$

$$\therefore 1 + 2qy^2 + 2qy = r^2y^3$$

$$\text{Let } y = x:$$

$$\therefore r^2x^3 - 2qx^2 - 2qx - 1 = 0$$

QUESTION 5:

(a) i $f(x) = -x^2 + 5x - 4$
 CURVE CUTS X AXIS WHEN $y = 0$

$$\therefore -x^2 + 5x - 4 = 0$$

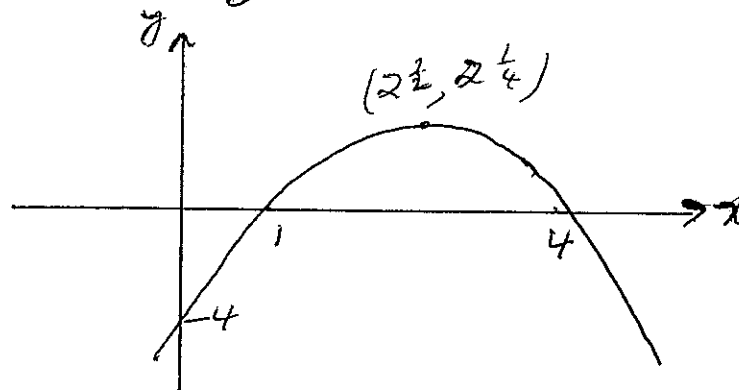
$$x^2 - 5x + 4 = 0$$

$$(x-4)(x-1) = 0$$

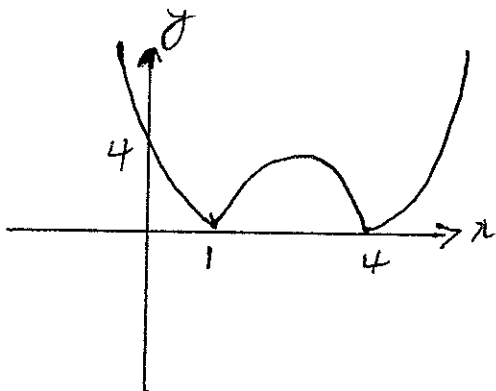
$$x = 1, 4$$

AXIS OF SYM IS $x = \frac{-5}{-2} = 2\frac{1}{2}$

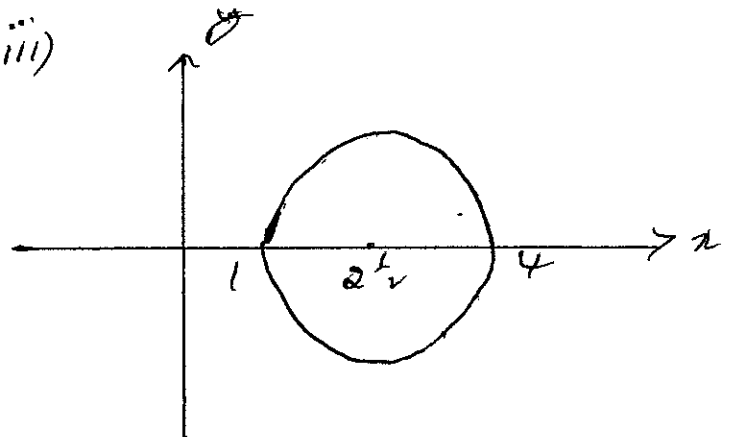
$$y = 2\frac{1}{4}$$



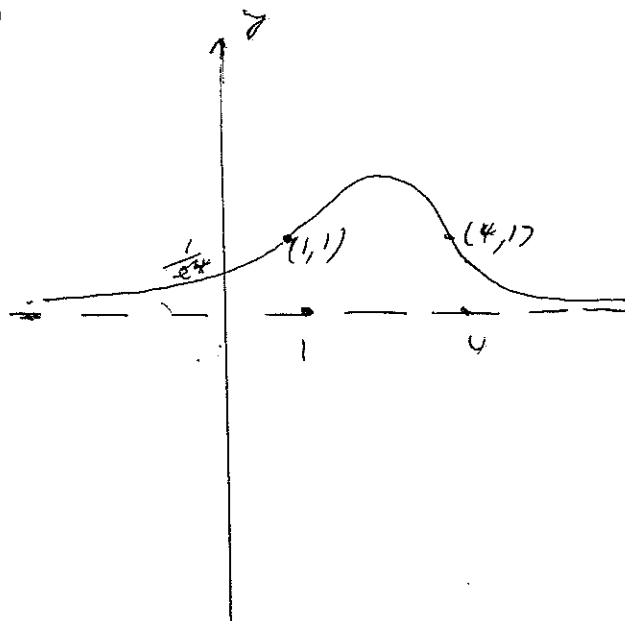
ii



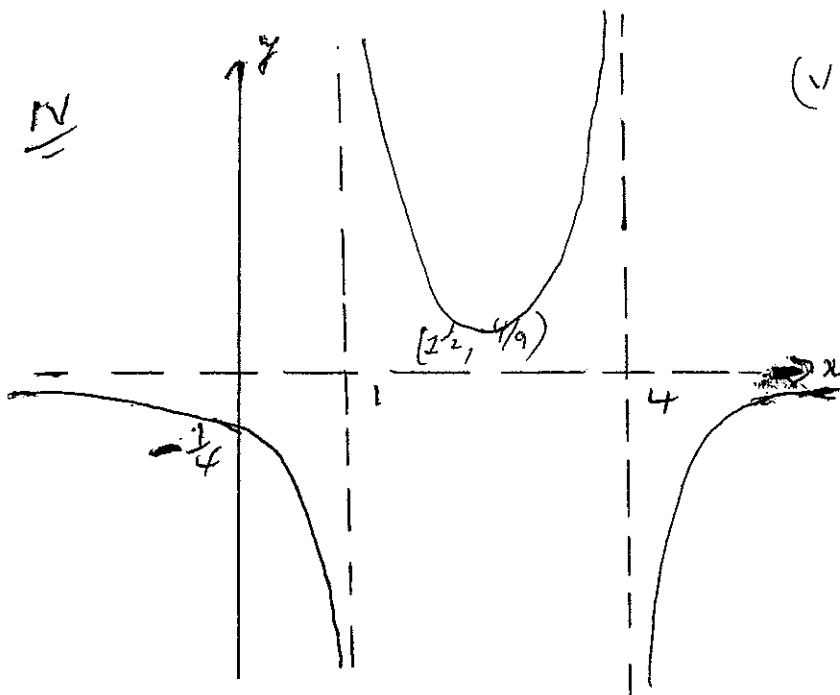
(iii)



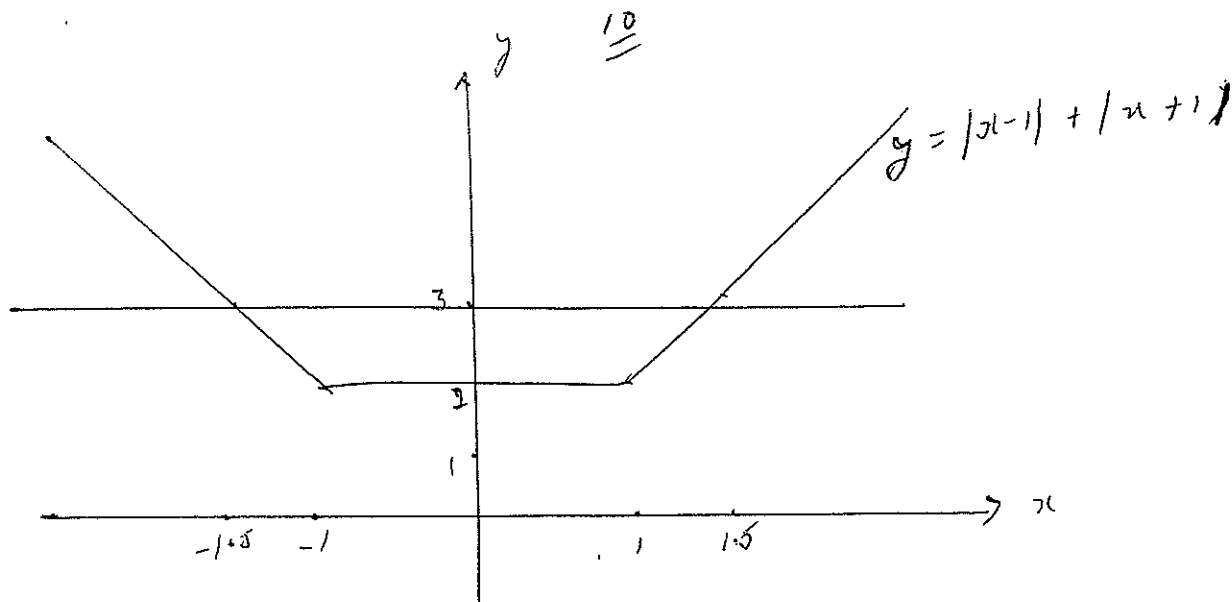
(iv)



iv



(b)



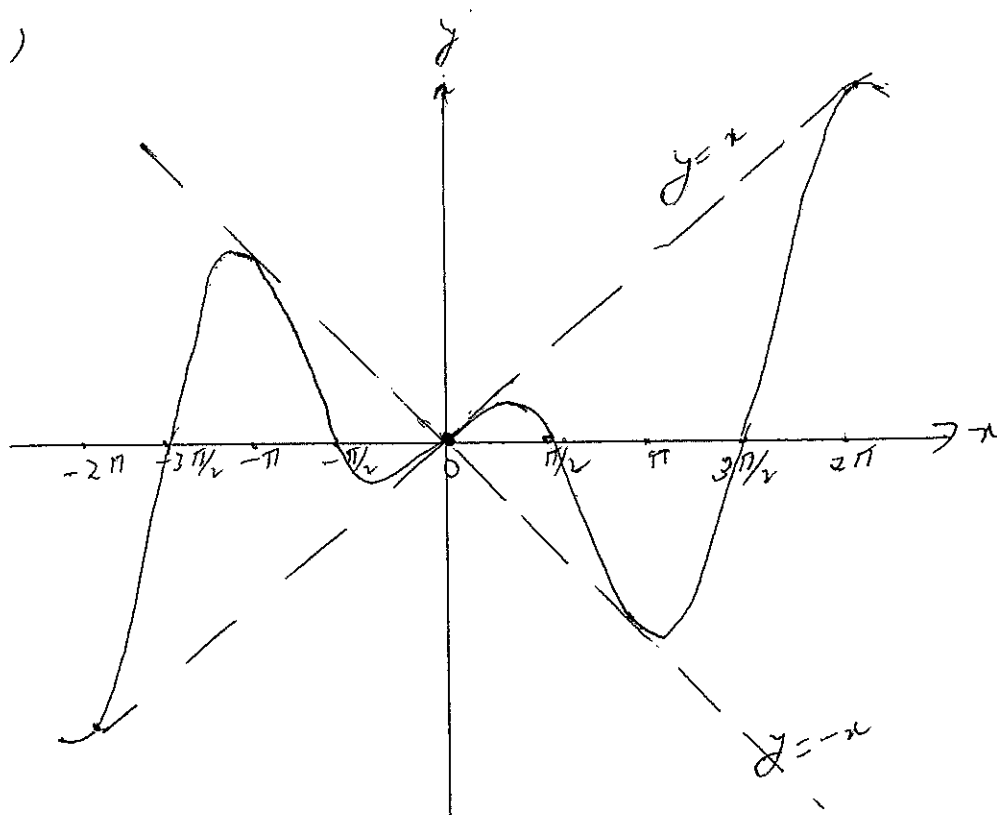
$$\begin{aligned} C1: \quad x-1 + x+1 &= 3 \\ 2x &= 3 \\ x &= 1\frac{1}{2} \end{aligned}$$

INEQUALITY HOLDS WHEN $x > 1\frac{1}{2}$

$$\begin{aligned} C2: \quad -x+1 - x-1 &= 3 \\ -2x &= 3 \\ x &= -1\frac{1}{2} \end{aligned}$$

INEQUALITY HOLDS WHEN $x < -1\frac{1}{2}$

(c)



$$\begin{aligned} f(x) &= x \cos x \\ f(-x) &= -x \cos(-x) \\ &= -x \cos x \\ &= -f(x) \end{aligned}$$

$\therefore$ ODD FUNCTION
 $\therefore$ POINT SYMMETRY ABOUT O.

QUESTION 6:

$$\frac{x^2}{4} - \frac{y^2}{12} = 1$$

(i) $a = 2$, $b = \sqrt{12} = 2\sqrt{3}$

$$b^2 = a^2 (e^2 - 1)$$

$$12 = 4 (e^2 - 1)$$

$$4e^2 = 16$$

$$e^2 = 4, e = 2$$

$\therefore$ FOCI $(\pm ae, 0) = (\pm 4, 0)$

(ii) DIRECTRICES: $x = \pm \frac{a}{e}$
 $x = \pm 1$

(iii) ASYMPTOTES: $y = \pm \frac{b}{a} x$
 $= \pm \frac{2\sqrt{3}}{2} x$
 $= \pm \sqrt{3} x$

(iv) GRAD OF TAN = $\frac{dy}{dx}$

$$\frac{x}{2} - \frac{y}{6} \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = \frac{3x}{y}$$

at $(2 \sec \theta, 2\sqrt{3} \tan \theta)$ $\frac{dy}{dx} = \frac{6 \sec \theta}{2\sqrt{3} \tan \theta}$
 $= \frac{3}{\cos \theta} \div \frac{\sqrt{3} \sin \theta}{\cos \theta}$
 $= \frac{\sqrt{3}}{\sin \theta}$

$\therefore$ EQUⁿ OF TANGENT IS
 $y - 2\sqrt{3} \tan \theta = \frac{\sqrt{3}}{\sin \theta} (x - 2 \sec \theta)$

$$\sin \theta y - \frac{2\sqrt{3} \sin^2 \theta}{\cos \theta} = \sqrt{3} x - \frac{2\sqrt{3}}{\cos \theta}$$

$\therefore \cos \theta \sin \theta y - 2\sqrt{3} \sin^2 \theta = \sqrt{3} x \cos \theta - 2\sqrt{3}$
 $\cos \theta \sin \theta y - 2\sqrt{3} (1 - \cos^2 \theta) = \sqrt{3} x \cos \theta - 2\sqrt{3}$

$\therefore \sqrt{3} x \cos \theta - \cos \theta \sin \theta y = 2\sqrt{3} \cos^2 \theta$

$$\frac{\sqrt{3} x}{\cos \theta} - \frac{\sin \theta y}{\cos \theta} = 2\sqrt{3}$$

$\therefore \frac{x \sec \theta}{2} - \frac{y \tan \theta}{2\sqrt{3}} = 1$

$$(V) \cdot \frac{x \sec \theta}{2} - \frac{y \tan \theta}{2\sqrt{3}} = 1 \quad \text{--- (1)}$$

$$x = 1 \quad \text{--- (2)}$$

$$\therefore \frac{\sec \theta}{2} - \frac{y \tan \theta}{2\sqrt{3}} = 1$$

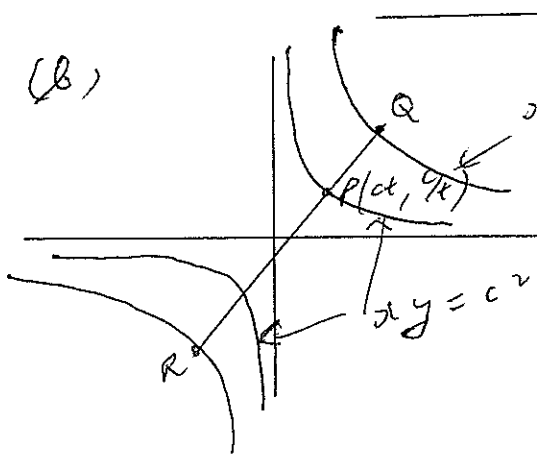
$$\therefore \sqrt{3} \sec \theta - y \tan \theta = 2\sqrt{3}$$

$$\frac{\sqrt{3}}{\cos \theta} - \frac{y \sin \theta}{\cos \theta} = 2\sqrt{3}$$

$$\therefore y \sin \theta = \sqrt{3} - 2\sqrt{3} \cos \theta$$

$$y = \frac{\sqrt{3} - 2\sqrt{3} \cos \theta}{\sin \theta} \times \frac{2\sqrt{3}}{2\sqrt{3}}$$

$$= \frac{6 - 12 \cos \theta}{2\sqrt{3} \sin \theta}$$



$$xy = c^2 \quad \perp \quad \text{GRAD OF TAN} = \frac{dy}{dx}$$

$$x \frac{dy}{dx} + y = 0$$

$$\therefore \frac{dy}{dx} = -\frac{y}{x}$$

$$\text{at } P \quad = -c/t / ct$$

$$= -\frac{1}{t^2}$$

$$\therefore \text{GRAD OF } \perp = t^2$$

$$\therefore \text{EQU OF } \perp \text{ IS}$$

$$y - \frac{c}{t} = t^2 (x - ct)$$

$$ty - c = t^3 x - ct^4 + c$$

$$ty = t^3 x - ct^4 + c$$

$$(11) \quad ty = t^3 x - ct^4 + c \quad \text{--- (1)}$$

$$x^2 - y^2 = a^2 \quad \text{--- (2)}$$

$$\therefore x^2 - \left(\frac{t^3 x - ct^4 + c}{t} \right)^2 = a^2$$

$$\therefore t^2 x^2 - [t^6 x^2 - ct^7 x + ct^3 x - ct^7 x + c^2 t^8 - c^2 t^4 + ct^3 x - c^2 t^2 + c^2] = a^2 + t^2$$

$$\therefore x^2 (t^2 - t^6) + x (2ct^7 - 2ct^3) - c^2 t^8 + 2c^2 t^4 + c^2 - a^2 t^2 = 0$$

THE SOLUTIONS TO THIS EQUATION PRODUCES THE COORDINATES OF Q & R. IF P IS THE MIDPOINT THEN ITS X COORDINATE = $\frac{Q_x + R_x}{2}$ i.e. SUM OF ROOTS

$$\text{SUM OF ROOTS} = -\frac{b}{a} = -\frac{(2ct^7 - 2ct^3)}{t^2 - t^6}$$

$$= \frac{2ct^3 (-t^4 + 1)}{t^2 (1 - t^4)}$$

$$= 2ct$$

$$= x \text{ COORDINATE OF } P$$

$\therefore P$ IS THE MIDPOINT OF QR.

QUESTION 7:

15

$$(a) \quad \perp \quad (x-y)^2 \geq 0$$

$$x^2 - 2xy + y^2 \geq 0$$

$$x^2 + y^2 \geq 2xy$$

$$\begin{aligned} \text{ii} \quad \frac{x}{y} + \frac{y}{x} &= \frac{x^2 + y^2}{xy} \\ &\geq \frac{x^2 + y^2}{\frac{x^2 + y^2}{2}} \end{aligned}$$

$$\geq 2$$

$$\text{iii} \quad x^3 + y^3 = (x+y)(x^2 - xy + y^2)$$

$$\text{But } x^2 + y^2 \geq 2xy$$

$$\therefore x^2 - xy + y^2 \geq xy$$

$$\begin{aligned} \therefore x^3 + y^3 &\geq (x+y)xy \\ &\geq \left(\frac{x}{2} + \frac{y}{2}\right)xy \cdot 2 \\ &\geq 2xyz \end{aligned}$$

$$\text{SIMILARLY: } x^3 + z^3 \geq 2xyz$$

$$\text{and } y^3 + z^3 \geq 2xyz$$

$$\therefore 2x^3 + 2y^3 + 2z^3 \geq 6xyz$$

$$\therefore x^3 + y^3 + z^3 \geq 3xyz$$

(b) Construct AB
 $\angle RPB = \angle AQB$ (EXT L OF CYCLIC QUAD
 $=$ INT OPP L)
 $\angle AQB = \angle BQS$ (L's IN SAME SEGMENT
 OF CIRCLE ARE EQUAL)

$$\therefore \angle RPB = \angle BQS$$

$\therefore MPBQ$ IS A CYCLIC QUAD AS
 EXTERIOR ANGLE ($\angle BQS$) EQUALS INTERIOR
 OPPOSITE ANGLE ($\angle RPB$).

$$\begin{aligned}
 \text{"(c)."} \quad \text{L.H.S.} &= \cos^6 \theta - \sin^6 \theta \stackrel{14}{=} \\
 &= (\cos^2 \theta - \sin^2 \theta)(\cos^4 \theta + \cos^2 \theta \sin^2 \theta + \sin^4 \theta) \\
 &= \cos 2\theta \left[(\cos^2 \theta + \sin^2 \theta)^2 - \cos^2 \theta \sin^2 \theta \right] \\
 &= \cos 2\theta \left[1 - \left(\frac{2 \cos \theta \sin \theta}{2} \right)^2 \right] \\
 &= \cos 2\theta \left[1 - \frac{1}{4} \sin^2 2\theta \right] \\
 &= \text{R.H.S.}
 \end{aligned}$$

QUESTION 8:

QUESTION 8:

15

$$(1) \quad \vec{BTX} = \vec{BPO} \quad (\text{CORR. } \angle \text{ TO } TX \parallel OP)$$

$$\vec{TXB} = \vec{POB} \quad (90^\circ)$$

$$\therefore \Delta TXB \parallel \Delta POB$$

$$\therefore \frac{TX}{a} = \frac{h}{r}$$

$$\therefore TX = \frac{ah}{r}$$

(1) (ONE SIDE OF RECTANGLE ZTXY)

$$\Delta PZT \parallel \Delta PAB$$

$$\therefore \frac{ZT}{PT} = \frac{AB}{PB}$$

$$\therefore \frac{ZT}{\sqrt{h^2 + r^2} - \sqrt{\frac{a^2 h^2}{r^2} + a^2}} = \frac{b}{\sqrt{h^2 + r^2}}$$

$$\therefore ZT = \frac{b}{\sqrt{h^2 + r^2} - \sqrt{\frac{a^2 h^2}{r^2} + a^2}} \times \left\{ \sqrt{h^2 + r^2} - \sqrt{\frac{a^2 h^2}{r^2} + a^2} \right\}$$

$$= \frac{b}{\sqrt{h^2 + r^2}} \times \left\{ \sqrt{h^2 + r^2} - \frac{a \sqrt{h^2 + r^2}}{r} \right\}$$

$$= \frac{b}{\sqrt{h^2 + r^2}} \times \frac{\sqrt{h^2 + r^2} (r - a)}{r}$$

$$= \frac{b(r - a)}{r} \quad (2)$$

$$\therefore \text{Volume of } S \text{ is } \left(\frac{r-a}{r} \right) b \cdot TX \cdot ZT \cdot \delta a$$

$$= \left(\frac{r-a}{r} \right) b \left(\frac{ah}{r} \right) \delta a$$

$$(ii) \therefore V = \int_0^r \left(\frac{r-a}{r} \right) b \left(\frac{ah}{r} \right) da$$

$$= \frac{b}{r^2} \int_0^r rha - ha^2 da$$

$$= \frac{b}{r^2} \left[\frac{rha^2}{2} - \frac{ha^3}{3} \right]_0^r$$

$$= \frac{b}{r^2} \left\{ \frac{r^3 h}{2} - \frac{r^3 h}{3} \right\}$$

$$= \frac{b}{r^2} \times \frac{r^3 h}{6}$$

$$= \frac{1}{6} h b r$$

(1) (i) PROVE $e - S_n = e \int_0^1 \frac{x^n}{n!} e^{-x} dx$ 16

STEP 1: PROVE TRUE FOR $n=1$

$$\begin{aligned}
 \text{LHS} &= e - S_1 \\
 &= e - \left(1 + \sum_{r=1}^1 \frac{1}{r!}\right) \\
 &= e - (1+1) \\
 &= e - 2
 \end{aligned}
 \qquad
 \begin{aligned}
 \text{RHS} &= e \int_0^1 \frac{x^1}{1!} e^{-x} dx \\
 &= e \int_0^1 x e^{-x} dx \\
 &= e \left\{ \left[x x e^{-x} \right]_0^1 - \int_0^1 -e^{-x} x^1 dx \right\} \\
 &= e \left\{ -\frac{1}{e} + \int_0^1 e^{-x} dx \right\} \\
 &= e \left\{ -\frac{1}{e} + \left[-e^{-x} \right]_0^1 \right\} \\
 &= e \left\{ -\frac{1}{e} + \left(-\frac{1}{e} - -e^0 \right) \right\} \\
 &= e \left\{ -\frac{1}{e} - \frac{1}{e} + 1 \right\} \\
 &= e - 2 = \text{LHS}
 \end{aligned}$$

$\therefore$ TRUE FOR $n=1$

STEP 2: ASSUME TRUE FOR $n=k$
 i.e. $e - S_k = e \int_0^1 \frac{x^k}{k!} e^{-x} dx$

$$\text{i.e. } S_k = e - e \int_0^1 \frac{x^k}{k!} e^{-x} dx$$

STEP 3: PROVE TRUE FOR $n=k+1$
 i.e. AIM TO PROVE: $e - S_{k+1} = e \int_0^1 \frac{x^{k+1}}{(k+1)!} e^{-x} dx$
 i.e. $S_{k+1} = e - e \int_0^1 \frac{x^{k+1}}{(k+1)!} e^{-x} dx$

$$\begin{aligned}
 \text{NOW } S_{k+1} &= S_k + T_{k+1} \\
 &= e - e \int_0^1 \frac{x^k}{k!} e^{-x} dx + \frac{1}{(k+1)!} \\
 &= e - \frac{e}{k!} \left\{ \left[\frac{x^{k+1}}{k+1} e^{-x} \right]_0^1 - \int_0^1 \frac{x^{k+1}}{k+1} x e^{-x} dx \right\} + \frac{1}{(k+1)!} \\
 &= e - \frac{e}{k!} \left\{ \frac{1}{k+1} \cdot \frac{1}{e} + \int_0^1 \frac{x^{k+1}}{k+1} e^{-x} dx \right\} + \frac{1}{(k+1)!} \\
 &= e - \frac{1}{k!(k+1)} - \frac{e}{k!} \int_0^1 \frac{x^{k+1}}{k+1} e^{-x} dx + \frac{1}{(k+1)!} \\
 &= e - \frac{1}{(k+1)!} - e \int_0^1 \frac{x^{k+1}}{(k+1)!} e^{-x} dx + \frac{1}{(k+1)!} \\
 &= e - e \int_0^1 \frac{x^{k+1}}{(k+1)!} e^{-x} dx
 \end{aligned}$$

STEP 4: THEREFORE TRUE FOR $n=k+1$ IF TRUE FOR $n=k$; BUT IT IS TRUE FOR $n=1$ & THEREFORE TRUE FOR $n=2$ & THEREFORE FOR $n=3$ & SO ON. BY THE PROCESS OF MATHEMATICAL INDUCTION IT IS TRUE FOR ALL INTEGRAL VALUES OF n .

Q(8) (b) (i) THIS IS THE SUM OF A GEOMETRIC SEQUENCE

$$\therefore S_n = \frac{1 - (-x^2)^{2n+1}}{1 - (-x^2)} = \frac{1 + (x^2)^{2n+1}}{1 + x^2} = \frac{1 + x^{4n+2}}{1 + x^2}$$

AS $2n+1$ IS ODD

(ii) $1 + x^2 \geq 1$ AS $x^2 \geq 0$

AS $1 - x^2 + x^4 - \dots + x^{4n} = \frac{1 + x^{4n+2}}{1 + x^2} = \frac{1}{1 + x^2} + \frac{x^{4n+2}}{1 + x^2}$

THEN $\frac{1}{1 + x^2} \leq 1 - x^2 + x^4 - \dots + x^{4n}$ ($\frac{x^{4n+2}}{1 + x^2}$ IS POS)

ALSO $\frac{1}{1 + x^2} + \frac{x^{4n+2}}{1 + x^2} \leq \frac{1}{1 + x^2} + x^{4n+2}$

$\therefore \frac{1}{1 + x^2} \leq 1 - x^2 + x^4 - \dots + x^{4n} \leq \frac{1}{1 + x^2} + x^{4n+2}$

(iii) $\therefore \int_0^y \frac{1}{1 + x^2} dx \leq \int_0^y \frac{1 + x^{4n+2}}{1 + x^2} dx \leq \int_0^y \frac{1}{1 + x^2} + x^{4n+2} dx$

$\therefore \tan^{-1} y \leq y - \frac{y^3}{3} + \frac{y^5}{5} - \dots + \frac{y^{4n+1}}{4n+1} \leq \tan^{-1} y + \frac{y^{4n+3}}{4n+3}$

BUT IF $0 \leq y \leq 1$ THEN $y^{4n+3} \leq 1$

$\therefore \tan^{-1} y + \frac{y^{4n+3}}{4n+3} \leq \tan^{-1} y + \frac{1}{4n+3}$

$\therefore \tan^{-1} y \leq y - \frac{y^3}{3} + \frac{y^5}{5} - \dots + \frac{y^{4n+1}}{4n+1} \leq \tan^{-1} y + \frac{1}{4n+3}$

(iv) SUBSTITUTE $y = 1$, $n = 250$

$\therefore \tan^{-1} 1 \leq 1 - \frac{1}{3} + \frac{1}{5} - \dots + \frac{1}{1001} \leq \tan^{-1} 1 + \frac{1}{1003}$

$\therefore \frac{\pi}{4} < 1 - \frac{1}{3} + \frac{1}{5} - \dots + \frac{1}{1001} < \frac{\pi}{4} + \frac{1}{1003}$

↑
RATIONAL

↑
IRRATIONAL

$\therefore$ NO EQUALITY SIGN.

BUT $1003 > 1000 \therefore \frac{1}{1003} < \frac{1}{1000}$

$\therefore 0 < \left(1 - \frac{1}{3} + \frac{1}{5} - \dots + \frac{1}{1001}\right) - \frac{\pi}{4} < \frac{1}{1000}$

$\therefore 0 < \left(1 - \frac{1}{3} + \frac{1}{5} - \dots + \frac{1}{1001}\right) - \frac{\pi}{4} < 10^{-3}$

$$(v) \frac{x \sec \theta}{2} - \frac{y \tan \theta}{2\sqrt{3}} = 1 \quad \text{--- (1)}$$

$$x = 1 \quad \text{--- (2)}$$

$$\therefore \frac{\sec \theta}{2} - \frac{y \tan \theta}{2\sqrt{3}} = 1$$

$$\sqrt{3} \sec \theta - y \tan \theta = 2\sqrt{3}$$

$$\frac{\sqrt{3}}{\cos \theta} - \frac{y \sin \theta}{\cos \theta} = 2\sqrt{3}$$

$$y \sin \theta = \sqrt{3} - 2\sqrt{3} \cos \theta$$

$$\therefore y = \frac{\sqrt{3} - 2\sqrt{3} \cos \theta}{\sin \theta} \times \frac{2\sqrt{3}}{2\sqrt{3}}$$

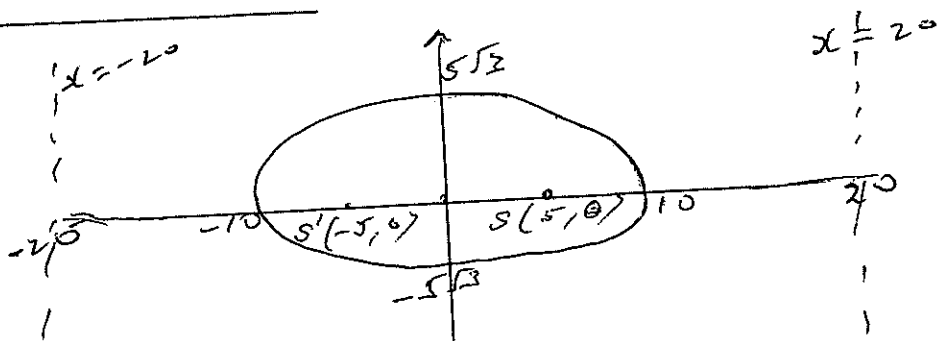
$$= \frac{6 - 12 \cos \theta}{2\sqrt{3} \sin \theta}$$

$$(b) \frac{x^2}{100} + \frac{y^2}{75} = 1$$

$$(i) \therefore a = 10, b = 5\sqrt{3}$$

$$b^2 = a^2 (1 - e^2)$$

$$\therefore e = \frac{1}{2}$$



$$(ii) \frac{2x}{100} + \frac{2y}{75} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{3x}{4y}$$

$$\text{AT } P(5, 7.5) \text{ grad of } \tan = -\frac{1}{2}$$

$$\therefore \text{grad of } \perp = 2$$

$$\therefore \text{EQU}^n \text{ OF } \perp \text{ is } y - 7.5 = 2(x - 5)$$

$$4x - 2y - 5 = 0$$

(iii) BY SYMMETRY THE INTERSECTION OF THE TWO NORMALS WILL BE ON THE X-AXIS & THIS WILL BE THE CENTRE OF THE CIRCLE

$$\therefore \text{WHEN } y = 0, x = 5/4 \therefore \text{CENTRE } (5/4, 0)$$

$$\text{RADIUS}^2 = (5 - 5/4)^2 + (7.5 - 0)^2$$

$$= \frac{1125}{16}$$

$$\therefore \text{EQU}^n \text{ OF CIRCLE IS } (x - \frac{5}{4})^2 + y^2 = \frac{1125}{16}$$

(11) Since $x^n \geq 0$ FOR $x \geq 0$ & $e^{-x} > 0$

$$e \int_0^1 \frac{x^n}{n!} e^{-x} dx = e \int_0^1 \text{FUNCTION IS ALWAYS POSITIVE} dx > 0$$

$$\therefore 0 < e - S_n$$

$$\begin{aligned} \text{Now } e \int_0^1 \frac{x^n}{n!} e^{-x} dx &< 3 \int_0^1 \frac{x^n}{n!} e^{-x} dx \quad \text{as } e < 3 \\ &< 3 \int_0^1 \frac{x^n}{n!} dx \quad \text{as } e^{-x} \leq 1 \text{ FOR } x \geq 0 \\ &< \frac{3}{n!} \left[\frac{x^{n+1}}{n+1} \right]_0^1 \\ &< \frac{3}{n!} \left\{ \frac{1}{n+1} \right\} \\ &< \frac{3}{(n+1)!} \end{aligned}$$

$$\therefore 0 < e - S_n < \frac{3}{(n+1)!}$$
